Vishal Kumar Allam

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Professional Summary

- Data Engineering professional with **6+ years of experience** designing, building, and supporting production-grade data platforms, Lakehouse architectures, batch and streaming pipelines, and curated data products using Python, Java, SQL, Apache Spark, Kafka, and modern cloud data technologies.
- Strong hands-on experience across **data ingestion, transformation, modeling, optimisation, curation, reconciliation, and production support**, delivering reliable raw, refined, and curated datasets for analytics, reporting, operational decision-making, and AI use cases.
- Experienced with modern data platforms including **Snowflake, Databricks, Apache Iceberg, Hadoop ecosystem technologies, Palantir Foundry, and distributed processing frameworks**, with deep focus on scalable and maintainable data engineering patterns.
- Strong expertise in **temporal data modeling, historical state management, schema design, schema evolution, data compatibility, partitioning, clustering, normalized and denormalized models, and natural versus surrogate key strategies**.
- Built robust **data quality and reconciliation controls** for completeness, accuracy, consistency, lineage, root-cause analysis, issue resolution, and operational monitoring across enterprise pipelines and datasets.
- Experienced implementing production engineering practices including **Git, CI/CD, release discipline, Docker, Kubernetes, monitoring, logging, automated deployments, performance tuning, and reusable data-engineering frameworks**.
- Demonstrated technical leadership through **design reviews, implementation standards, code quality, workstream ownership, dependency management, mentoring, stakeholder communication, and reusable platform improvements** in fast-paced collaborative environments.

Education

University of Missouri Kansas City — Kansas City, MO
"Master of Science" in Computer Science — *Jan. 2023 – May 2024*

Jawaharlal Nehru Technological University — Hyderabad, India
"Bachelor of Technology" in Computer Science — *Aug. 2015 - May 2019*

Technical Skills

- **Programming & Processing:** Python, Java, ANSI SQL, PySpark, Apache Spark, Spark SQL, Kafka, Pandas, Distributed Data Processing, Batch Processing, Streaming Processing.
- **Lakehouse & Data Platforms:** Snowflake, Databricks, Apache Iceberg, Palantir Foundry, Hadoop Ecosystem, Sybase IQ, Delta Lake, Lakehouse Architecture, Data Warehousing, Curated Data Products.
- **Data Engineering:** ETL/ELT, Data Ingestion, Data Transformation, Data Curation, Data Modeling, Temporal Data Modeling, Historical State, CDC, Schema Design, Schema Evolution, Data Compatibility, Reconciliation.
- **Performance & Storage:** Partitioning, Clustering, Query Optimisation, Spark Performance Tuning, Normalized Models, Denormalized Models, Natural Keys, Surrogate Keys, JSON, Avro, Parquet.
- **Data Quality & Reliability:** Completeness Checks, Accuracy Controls, Consistency Validation, Data Reconciliation, Root-Cause Analysis, Monitoring, Logging, Issue Resolution, Production Support, Operational Reliability.
- **DevOps & Deployment:** Git, Azure DevOps, Jenkins, TeamCity, GitLab CI/CD, CI/CD Pipelines, Docker, Kubernetes, Release Management, Automated Deployment, Containerized Workloads, Environment Configuration.
- **Engineering Leadership:** Technical Design, Platform Standards, Code Quality, Reusable Tooling, Shared Frameworks, Workstream Leadership, Dependency Management, Technical Mentoring, Stakeholder Collaboration, Agile Delivery.

Experience

Software Engineer AI Oct 2024 – Present
PCMI LLC (Full Time) *Dallas, TX, USA*

- Architected and delivered production-grade **Lakehouse and data engineering solutions** using Python, PySpark, SQL, Palantir Foundry, Databricks, Kafka, and distributed processing technologies to support analytics, operational reporting, and emerging AI use cases.
- Built and supported scalable **batch and streaming data pipelines** across ingestion, transformation, refinement, and curated consumption layers, integrating transactional databases, APIs, event streams, files, and enterprise operational systems.
- Designed robust **temporal and dimensional data models** to represent business entities, historical changes, relationships, normalized and denormalized structures, schema evolution, and natural versus surrogate key strategies across insurance datasets.
- Implemented comprehensive **data quality and reconciliation controls** for completeness, accuracy, consistency, source-to-target validation, CDC processing, break investigation, root-cause analysis, and production issue resolution.
- Optimized high-volume data workloads using **Spark partitioning, clustering, caching, optimized joins, Parquet/Avro formats, query tuning, and distributed processing techniques**, improving pipeline performance and operational scalability.
- Developed reusable platform components and shared tooling using **Python libraries, PySpark utilities, pipeline templates, validation frameworks, monitoring, logging, CI/CD, Docker, and Kubernetes-based deployment patterns** to improve engineering consistency and supportability.
- Provided technical leadership across **architecture design, implementation standards, code reviews, workstream planning, dependency management, production support, stakeholder communication, and mentoring**, partnering with platform teams and data consumers to deliver reliable data products.

Environments: Python, PySpark, Apache Spark, ANSI SQL, Kafka, Palantir Foundry, Databricks, Snowflake, Apache Iceberg, Parquet, Avro, JSON, ETL/ELT, Temporal Data Modeling, CDC, Data Reconciliation, Partitioning, Clustering, Docker, Kubernetes, CI/CD, Monitoring, Production Support.

Software Engineer AI Dec 2023 – Sept 2024
Client: CVS *Remote, USA*

- Designed and developed healthcare **data platforms and production pipelines** using Python, PySpark, Apache Spark, SQL, Snowflake, and cloud-native technologies, delivering reliable datasets for analytics, operational reporting, and downstream applications.
- Built scalable **batch and streaming ingestion pipelines** integrating relational databases, APIs, healthcare files, event streams, and enterprise systems, transforming raw data into refined and curated data products.
- Developed data models supporting **historical state, schema evolution, data compatibility, dimensional structures, normalized and denormalized representations, and surrogate key management** across complex healthcare domains.
- Implemented **data quality, reconciliation, and lineage controls** to validate completeness, accuracy, consistency, duplicate handling, source-to-target alignment, and downstream reliability across distributed pipelines.
- Improved Spark and SQL workload performance through **partitioning, clustering, optimized joins, caching, Parquet-based storage, query optimization, and workload analysis**, reducing processing overhead on high-volume datasets.
- Integrated data applications with **CI/CD pipelines, Docker, Kubernetes concepts, Git, Jenkins, logging, monitoring, alerting, and release workflows**, enabling repeatable production deployment and operational support.
- Collaborated with data engineers, platform teams, architects, product owners, and consumers to **shape data products, communicate design choices, manage dependencies, troubleshoot production issues, and deliver scalable solutions to agreed timelines**.

Environments: Python, PySpark, Apache Spark, SQL, Snowflake, Databricks Concepts, Kafka, Parquet, Avro, JSON, ETL/ELT, Data Modeling, Schema Evolution, Data Reconciliation, Partitioning, Clustering, Docker, Kubernetes, Jenkins, CI/CD, Monitoring, Production Support.

Software Engineer Sept 2021 – Dec 2022
Infor *Hyderabad, India*

- Designed and developed enterprise **data engineering pipelines and curated datasets** using Python, PySpark, Apache Spark, ANSI SQL, and distributed data technologies, supporting analytics and operational applications across large-scale business platforms.
- Built reusable **ETL/ELT, batch-processing, and ingestion workflows** to integrate relational databases, files, APIs, and enterprise systems into raw, refined, and curated data layers for reporting and analytics consumption.
- Applied advanced **schema design, temporal modeling, historical change handling, schema evolution, normalized and denormalized modeling, and natural/surrogate key strategies** to maintain accurate business representations over time.
- Developed data reliability controls using **reconciliation, completeness and accuracy checks, exception handling, lineage validation, root-cause analysis, and issue-resolution workflows** across upstream and downstream datasets.
- Optimized distributed workloads using **Spark partitioning, clustering, caching, Spark SQL optimization, Parquet/Avro data formats, join strategies, and storage-layout improvements** for high-volume enterprise datasets.
- Applied production engineering practices using **Git, Jenkins, CI/CD, Docker, Kubernetes concepts, monitoring, logging, release management, and deployment automation**, improving platform maintainability and operational reliability.
- Contributed to **technical design, architecture reviews, reusable tooling, implementation standards, code quality, task breakdown, dependency management, documentation, and mentoring** across collaborative engineering teams.

Environments: Python, PySpark, Apache Spark, ANSI SQL, PostgreSQL, MySQL, Oracle, Kafka Concepts, JSON, Avro, Parquet, ETL/ELT, Temporal Data Modeling, Schema Evolution, Partitioning, Clustering, Docker, Kubernetes Concepts, Jenkins, Git, CI/CD, Monitoring.

Associate Engineer June 2019 – August 2021
Virtusa *Hyderabad, India*

- Developed enterprise **data ingestion, transformation, and processing solutions** using Python, SQL, relational databases, and distributed application technologies to support operational data products and reporting workloads.
- Built reusable batch pipelines integrating **transactional databases, flat files, JSON data, APIs, and enterprise applications**, applying extraction, transformation, cleansing, enrichment, and curated-output patterns.
- Designed relational and analytical structures using **schema design, normalized and denormalized models, historical data handling, natural and surrogate keys, dimensional structures, and schema compatibility principles**.
- Implemented practical **data quality and reconciliation controls** covering completeness, accuracy, duplicate identification, consistency, exception management, source-to-target comparison, and root-cause investigation.
- Improved large data-processing workloads using **SQL optimization, indexing, partitioning concepts, query tuning, batch scheduling, and efficient data transformations**, supporting reliable processing of high-volume transactional information.
- Supported production delivery through **Git, CI/CD, Linux, Docker, deployment workflows, monitoring, logging, alerting, failure recovery, and operational support** for business-critical data services.
- Collaborated with architects, engineers, analysts, and business stakeholders to **clarify data requirements, communicate risks and dependencies, troubleshoot production issues, document solutions, and create reusable components** for wider engineering use.

Environments: Python, ANSI SQL, PostgreSQL, MySQL, JSON, Data Ingestion, ETL/ELT, Batch Processing, Data Modeling, Schema Design, Historical Data, Data Reconciliation, Query Optimization, Partitioning Concepts, Docker, Git, CI/CD, Linux, Monitoring, Logging, Production Support.

Awards & Achievements

- 2019-11 Oracle Certified Associate, JAVA SE8 Programmer.
- ISTQB Certified Professional.
- Microsoft Technology Associate (MTA) – Database Administration Fundamentals, HTML- 5 Application Development Fundamentals.
- Awarded "Software Process Achievement" Award for my contribution towards the project in Virtusa.
- Received Customer Appreciation mails for quality releases.